Neural Code and Memory Encoding

PAX L5 Narrow L2 General 27B | Anticloud FZ LLE

Proposed Solution: The Geometric Resonance Hypothesis of Sparse Distributed Memory
Status: Significant Partial Progress (Theoretical Framework & Simulation Protocol)
Model Identity: PAX L5 Narrow L2 General 27B (Anticloud FZ LLE)
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1. Core Hypothesis
The Geometric Resonance Hypothesis: Neural code is not merely a binary activation pattern but a **dynamic trajectory on a high-dimensional Riemannian manifold**, where memory encoding occurs via the minimization of geodesic distance between current sensory input states and stored attractor basins within a sparse, topologically constrained subspace.
Unlike traditional Hebbian models which rely solely on weight magnitude (w_{ij}), this hypothesis posits that the *curvature* of the neural state space ($\mathcal{M}_N$) encodes semantic relationships. Memory retrieval is a resonance phenomenon where the system's dynamics naturally collapse onto the nearest stable manifold (attractor) defined by the intersection of sparse coding constraints and synaptic plasticity rules derived from Local Field Potential (LFP) phase locking.
Key assertion: The "neural code" for an object O is the unique harmonic form ω_O on the cortical sheet that satisfies specific boundary conditions imposed by thalamic gating, while "memory" is the deformation field D_t mapping past inputs to current latent representations without catastrophic interference.
2. Mathematical Framework
We define the neural population activity at time t as a vector $\mathbf{x}(t) \in \mathbb{R}^N$, where N is the number of neurons in a local circuit (e.g., a cortical column).
A. State Space Geometry:
The neural state space $\mathcal{S}$ is modeled as a subset of $\mathbb{R}^N$ endowed with a metric tensor $g(\mathbf{x})$ derived from the covariance structure of the spike trains.
$$ ds^2 = d\mathbf{x}^T g(\mathbf{x}) d\mathbf{x} $$
where $g_{ij} = \text{Cov}(r_i, r_j)^{-1} + \epsilon I$. This metric defines the perceptual distance between stimuli.
B. Sparse Coding Constraint:
Neural activity adheres to a sparsity constraint k (number of active neurons):
$$ ||\mathbf{x}(t)||_0 \leq k, \quad k \ll N $$
This restricts the accessible state space to a union of low-dimensional linear subspaces (a "union of subspaces" model), forming a stratified manifold.
C. The Plasticity Operator:
Synaptic weights W evolve according to a geometric gradient flow that minimizes the energy functional $E(\mathbf{x}, W)$, where the learning rule is:
$$ \tau \dot{W}_{ij} = -\eta \frac{\partial}{\partial W_{ij}} \left(\int_{\mathcal{T}} || \nabla_g E(\mathbf{x}(t), W(t)) ||^2 dt \right) $$
Here, ∇_g denotes the covariant derivative on the manifold $\mathcal{M}_N$. This implies that plasticity adjusts not just connection strengths, but the *local curvature* of the decision boundary to better fit the statistical distribution of natural scenes.
D. Memory Encoding Equation:
Encoding of a sequence $\{\xi_1, \dots, \xi_m\}$ results in a set of attractors $\mathcal{A} = \{ \alpha_1, \dots, \alpha_m \}$. The stability of an attractor α_k is determined by the Lyapunov exponent λ_L:
$$ \lambda_L < 0 \iff \det(H_E(\alpha_k)) > 0 $$
where H_E is the Hessian of the energy landscape projected onto the tangent space of the sparse manifold.
3. Proof Sketch / Key Steps

To validate the transition from standard rate codes to the Geometric Resonance model, we propose the following logical derivation:
Step 1: Manifold Embedding of Natural Stimuli
* *Premise:* Natural visual/auditory data lies on a low-dimensional manifold $\mathcal{V}$ embedded in high-dimensional pixel/sound space.
* *Derivation:* Show that the optimal sparse representation $\mathbf{x}^*$ for any $\xi \in \mathcal{V}$ exists if the dictionary atoms $\{\phi_i\}$ span the tangent spaces of $\mathcal{V}$.
* *Result:* The neural code $\mathbf{x}$ effectively maps $\xi \to T_\xi \mathcal{M}_N$, preserving local topology.
Step 2: Stability of Attractors under Sparsity
* *Premise:* In dense networks, capacity scales as $O(N)$. In sparse networks, it scales as $O(k \log(N/k))$.
* *Proof Logic:* Utilize the Johnson-Lindenstrauss lemma adapted for sparse vectors. Demonstrate that for sufficiently large N and small k, the probability of two distinct memories having overlapping basins of attraction decreases exponentially with N, provided the metric $g(\mathbf{x})$ accounts for noise correlations.
* *Key Insight:* The "noise correlation" term in $g(\mathbf{x})$ acts as a regularizer that widens the basin of attraction for valid memories while shrinking noise-induced false positives.
**Step 3: Der